

$$1.1) \operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4} (|\vec{u} + \vec{v}|^2 - |\vec{u} - \vec{v}|^2)$$

$$\text{DŮKAZ: } |\vec{u} + \vec{v}|^2 - |\vec{u} - \vec{v}|^2 = \langle \vec{u} + \vec{v}, \vec{u} + \vec{v} \rangle - \langle \vec{u} - \vec{v}, \vec{u} - \vec{v} \rangle =$$

$$\langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, \vec{v} \rangle + \langle \vec{v}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle - (\langle \vec{u}, \vec{u} \rangle - \langle \vec{u}, \vec{v} \rangle - \langle \vec{v}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle) =$$

$$2\langle \vec{u}, \vec{v} \rangle + 2\langle \vec{v}, \vec{u} \rangle = 2(\langle \vec{u}, \vec{v} \rangle + \overline{\langle \vec{u}, \vec{v} \rangle}) = \quad x + \bar{x} = 2\operatorname{Re}$$

$$2(2\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle)) = 4\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) \quad \dots \text{po vylíčení 4 rovnosti platí}$$

$$\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4} (|\vec{u} - i\vec{v}|^2 - |\vec{u} + i\vec{v}|^2)$$

$$\text{DŮKAZ: } |\vec{u} - i\vec{v}|^2 - |\vec{u} + i\vec{v}|^2 = \langle \vec{u} - i\vec{v}, \vec{u} - i\vec{v} \rangle - \langle \vec{u} + i\vec{v}, \vec{u} + i\vec{v} \rangle =$$

$$\langle \vec{u}, \vec{u} \rangle - \langle \vec{u}, i\vec{v} \rangle - \langle i\vec{v}, \vec{u} \rangle + \langle i\vec{v}, i\vec{v} \rangle - (\langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, i\vec{v} \rangle + \langle i\vec{v}, \vec{u} \rangle + \langle i\vec{v}, i\vec{v} \rangle) = 2(-i)\langle \vec{u}, \vec{v} \rangle + i\langle \vec{v}, \vec{u} \rangle =$$

$$2(-i)(\langle \vec{u}, \vec{v} \rangle - \overline{\langle \vec{u}, \vec{v} \rangle}) =$$

$$x - \bar{x} = 2i\operatorname{Im}$$

$$2(-i)2i\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) = 4\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) \quad \dots \text{po vylíčení 4 rovnosti platí}$$

1.2) Dokažte, že platí všechny axiomy a definice skalárního součinu

$$(1) \text{ Symetrie: } \langle \vec{u}, \vec{v} \rangle = \vec{u}^T A \vec{v} = (\vec{u}^T A \vec{v})^T = \vec{v}^T A^T \vec{u} = \vec{v}^T A \vec{u} = \langle \vec{v}, \vec{u} \rangle$$

$$(2) \text{ Linearita: } \langle \vec{u}, A \cdot \vec{v} \rangle = \vec{u}^T A \cdot A \cdot \vec{v} = A(\vec{u}^T A \vec{v}) = A \cdot \langle \vec{u}, \vec{v} \rangle$$

$$\langle \vec{u}, \vec{v} + \vec{w} \rangle = \vec{u}^T A(\vec{v} + \vec{w}) = \vec{u}^T A \vec{v} + \vec{u}^T A \vec{w} = \langle \vec{u}, \vec{v} \rangle + \langle \vec{u}, \vec{w} \rangle$$